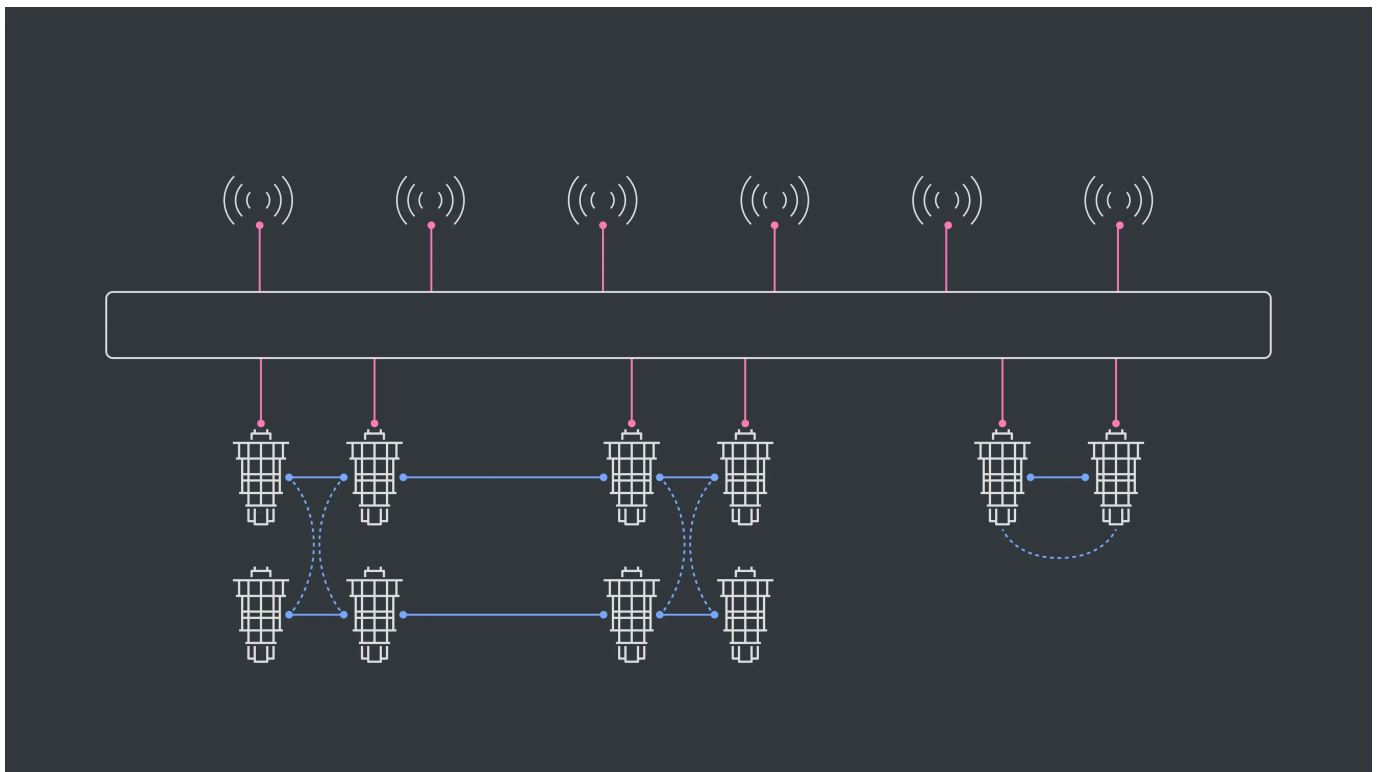


Quantum Research

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Scaling beyond our roadmap with networked quantum computers

Initiating the research and collaboration needed to build the quantum computers of the future.



Date

20 Nov 2025

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framework to compute with CPUs, GPUs, and QPUs together. Yet for scaling beyond our roadmap, the critical connective tissue for this vision will be networking quantum computers.

As with classical computing prior, the future of quantum-centric supercomputing requires industry-wide partnerships among a variety of vendors and research organizations, together contributing hardware, software, and knowledge. We capture our plan to build this future in our development roadmap, which charts our progress to build a computer capable of running quantum circuits with one billion operations on 2,000 qubits by 2033. However, scaling circuits to further orders of magnitude in both the numbers of operations and across qubits will require distributed quantum computing with connected systems. Beyond that, we hope to realize a quantum computing internet.

Therefore, we're kicking off research and initiating collaborations required by this future. Earlier this month, we announced that we'd be collaborating with [four of the five NQISR centers](#), including work focused on accelerating research for technologies beyond our development roadmap—like the quantum computing internet. Today, we also announced our plans with Cisco to study how we can link quantum processors to lay the groundwork for distributed quantum computing.

Networking quantum computers poses a grand challenge requiring groundbreaking research and development, which requires a step-wise approach to solve. Our first milestone will be entangling a pair of cryogenically separated quantum processors within the next five years. And this will only be possible with our partners.

Networking quantum computers

Understanding quantum networking requires understanding classical networking first. Computers encode information in binary code and process it by applying instructions which alter the code. You can give these instructions directly to the chip, or send them as messages to another chip with wired and wireless links. Classical computers follow the rules of cause-and-effect; when two computers are linked, one waits for a message and then acts upon message receipt. If you connect lots of



Quantum computers encode information into quantum states, and process using the often-counterintuitive mathematics that nature follows at the atomic scale. Like today's computers, you can imagine linking quantum computer nodes into quantum compute clusters or distributed networks. You could further extend this quantum network into a larger-scale quantum computing internet. We require quantum networks for the fullest realization of quantum computation.

Quantum and classical networks have profound differences. When you open a quantum link between objects, you generate entanglement between them. That means they no longer follow the classical rules of cause-and-effect; they act as a single entity, mathematically. Two quantum processors entangled over a quantum link, even at long distances, act like a single quantum computer. Operations applied at one node could instantly alter the outcomes measured at the nodes it's entangled with. Quantum computers (and quantum links) only have quantum properties during the calculation—at the end of your calculation, the quantum information collapses into a single classical output, a binary string, one per node.

This doesn't mean information travels faster than light. There's a "quantum no-communication theorem" that essentially says that there's nothing you can do at node A that will instantly be meaningful at node B after you collapse the quantum information to measure the outputs. If we spread information across both nodes, then understanding the whole story still requires gathering the outputs from the other nodes and communicating them over a classical link.

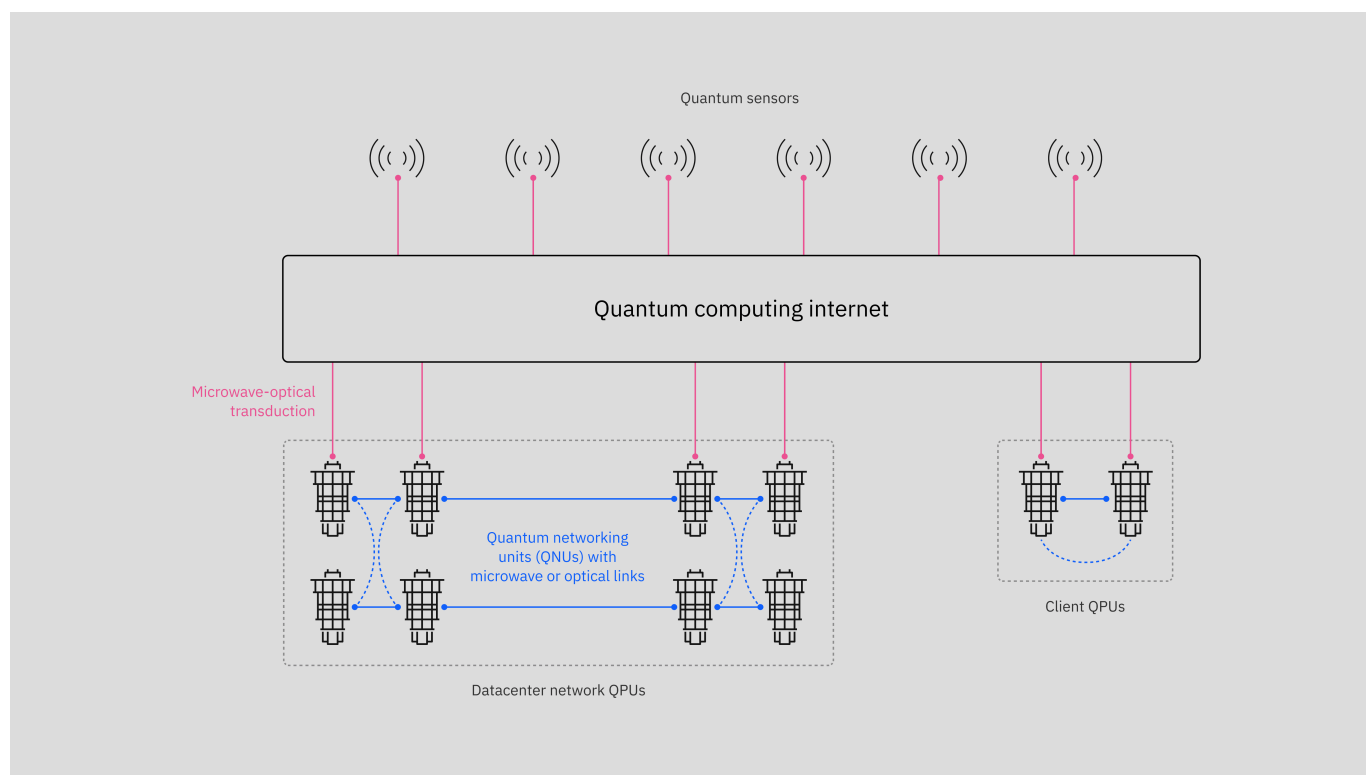
So, what's the purpose of networked quantum computers? Links would offer us the ability to build larger quantum datacenters, where we connect many quantum computers to achieve higher qubit counts and larger quantum circuits. They could further augment quantum sensing—systems that use quantum principles to make high-precision measurement. Astronomy labs like the Laser Interferometer Gravitational-Wave Observatory are already using quantum sensors to perform ultra-precise measurements in search of gravitational waves from colliding black holes. A quantum link to a quantum sensor could perhaps let us create a network of sensors to increase their precision through a technique called interferometry.



Quantum links, networks, and even a quantum computing internet all play a central role in our vision for the future of computing.

At the processing level, our goal for running ever-larger quantum computations requires linking modular quantum processors over a short range, which we have demonstrated with our Crossbill and Flamingo processors. But longer-range links, would let us link quantum processing units at the scale of meters to create a quantum computing cluster in a datacenter.

At the heart of this linking is the quantum networking unit, or QNU. QNUs are the interfaces between the processors and interconnects—they translate static qubits encoded on stationary processors into “flying” qubits that can propagate and travel across a network. Photons are the natural element to achieve flying qubits, but the specific frequency of these photons, be it optical or microwave, could define the type of infrastructure over which we imagine this network.



We've already started collaborating with government, academia, and business partners to begin researching and developing the components required to build distributed quantum computers, and in the future, a quantum computing internet.



collaborators.

- Internally, we're developing l-couplers, designed to operate inside dilution refrigerators at the same temperature as our quantum processors and connect QPUs on the one meter scale. We must pursue high-fidelity l-couplers to realize our goal for Starling and delivering fault-tolerant quantum computing in 2029.
- In partnership with Fermilab's Superconducting Quantum Materials and Systems Center, we're exploring connectors at the one- to ten-meter scale, designed to pass information at slightly higher temperatures to link quantum computers in the same building to help realize the quantum data center.
- Finally—and perhaps the most challenging to realize—couplers designed to transmit information over kilometers. These couplers require a transducer, or a device that converts the energy of our photons from the microwave scale to the optical scale so we can send them over the link, plus further peripherals that allow us to generate entanglement over the link.

Today, we announced a planned partnership with Cisco to explore that final, most challenging piece— transducers and optical links between QPUs—as we work to realize the most challenging facet of our quantum computing internet vision. And once we build QNUs capable of linking QPUs over short and long distances, we can realize a true quantum computing internet—with QPUs networked across kilometers, in some cases working alongside quantum sensors.

Looking ahead

Quantum computers are already providing signals of value as the community realizes quantum advantage and develops new algorithms. However, we're only scratching the surface of what's possible with quantum computing. A full realization of this technology requires a quantum computing internet. It'll be a challenging journey, but we feel confident we can succeed with the help of our clients and partners.

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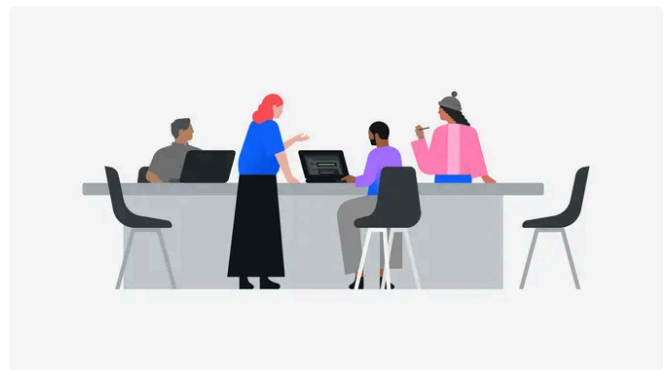
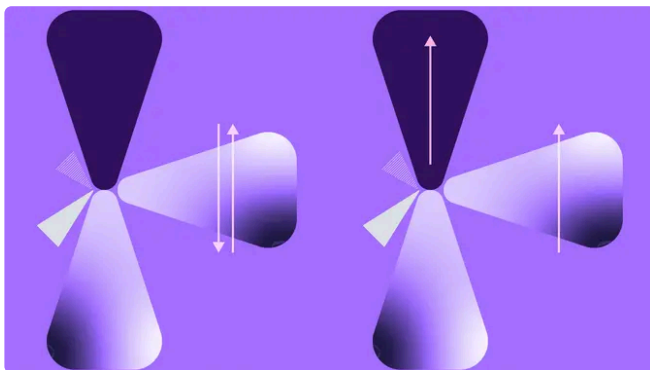
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